



KUWAIT UNIVERSITY
Faculty of Science
Department of Mathematics

Math 111: Linear Algebra

Final Exam, 15 May 2019

Duration: 120 minutes

- In questions 1-9, show all your work.
- Calculators and communication devices are not allowed in the examination room.

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (10 pts)	
Q2 (10 pts)	
Q3 (10 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (10 pts)	
Q7 (10 pts)	
Q8 (10 pts)	
Q9 (20 pts)	
Total	

1. [5 + 2.5 + 2.5 = 10 pts.] Let $A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix}$.

(a) Find the reduced row echelon form of A .

(b) Find a basis for the **row space** of A .

(c) Find a basis for the **column space** of A .

2. [5 + 5 = 10 pts.] Let $W = \{(a, a + b, a - b, b) : a, b \in \mathbb{R}\}$.

(a) Show that W is a subspace of $\mathbb{R}^4$.

(b) Find a basis for W , and $\dim(W)$.

3. [10 pts.] Show that if U and V are unit vectors in $\mathbb{R}^3$, then $\|U \times V\|^2 + (U \cdot V)^2 = 1$.

4. [2.5 × 4 = 10 pts.] Let A be an nonsingular 5×5 matrix.

(a) What is the **nullity** of A ?

(b) What is the **rank** of A ?

(c) Is 0 an eigenvalue of A ?

(d) Do the columns vectors of A form a basis for $\mathbb{R}^5$?

5. [10 pts.] Show that if λ is an eigenvalue of an invertible matrix A and X is a corresponding eigenvector, then λ^{-1} is an eigenvalue of A^{-1} and X is a corresponding eigenvector.

6. [10 pts.] Find the volume of the parallelepiped with vertex at the origin and edges $U = (2, -1, 0)$, $V = (1, -2, -2)$ and $W = (3, -1, 1)$.

7. [5 + 5 = 10 pts.] Let $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 3 & 2 & 0 & 0 & 0 \\ 1 & 5 & 3 & 0 & 0 \\ 4 & 6 & 7 & 4 & 0 \\ 9 & 2 & 1 & 2 & 5 \end{bmatrix}$.

(a) Find the cofactor A_{11} of the element $a_{11} = 1$.

(b) Find the eigenvalues of A^{-1} .

8. [10 pts.] Find parametric equations for the line in $\mathbb{R}^3$ passing through the point $P(-2, 5, -3)$ and orthogonal to the plane $2x - 3y + 4z + 7 = 0$.

9. [5 + 10 + 5 = 20 pts.] Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 3 & -1 \\ 3 & 0 & 0 \end{bmatrix}$.

(a) Show that the eigenvalues of A are $\lambda_1 = \lambda_2 = 3, \lambda_3 = -2$.

(b) For each eigenvalue λ , find the **rank** of the matrix $(\lambda I_3 - A)$.

(c) If possible, find a nonsingular matrix P and a diagonal matrix D such that $P^{-1}AP = D$.

1. [5 + 2.5 + 2.5 = 10 pts.] Let $A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix}$.

(a) Find the reduced row echelon form of A .

The reduced row echelon form of A is $\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(b) Find a basis for the **row space** of A .

$\{(1, 0, 0, 3), (0, 1, 0, 0), (0, 0, 1, 0)\}$.

(c) Find a basis for the **column space** of A .

$\{(1, 1, 0, 0), (1, 1, 0, 1), (2, 3, 1, 2)\}$.

2. [5 + 5 = 10 pts.] Let $W = \{(a, a + b, a - b, b) : a, b \in \mathbb{R}\}$.

(a) Show that W is a subspace of $\mathbb{R}^4$.

Since $(a, a + b, a - b, b) = a(1, 1, 1, 0) + b(0, 1, -1, 1)$, therefore $W = \text{span}\{(1, 1, 1, 0), (0, 1, -1, 1)\}$.
Thus, W is a subspace of $\mathbb{R}^4$.

(b) Find a basis for W , and $\dim(W)$.

Since $(1, 1, 1, 0)$ and $(0, 1, -1, 1)$ are linearly independent, therefore, they form a basis for W ,
and $\dim(W) = 2$.

3. [10 pts.] Show that if U and V are unit vectors in $\mathbb{R}^3$, then $\|U \times V\|^2 + (U \cdot V)^2 = 1$.

$\|U \times V\|^2 + (U \cdot V)^2 = \|U\|^2 \|V\|^2 \sin^2 \theta + \|U\|^2 \|V\|^2 \cos^2 \theta = \|U\|^2 \|V\|^2 [\sin^2 \theta + \cos^2 \theta] = \|U\|^2 \|V\|^2 = 1$.

4. [2.5 × 4 = 10 pts.] Let A be a nonsingular 5×5 matrix.

(a) What is the **nullity** of A ? 0.

(b) What is the **rank** of A ? 5.

(c) Is 0 an eigenvalue of A ? No.

(d) Do the columns vectors of A form a basis for $\mathbb{R}^5$? Yes.

5. [10 pts.] Show that if λ is an eigenvalue of an invertible matrix A and X is a corresponding eigenvector, then λ^{-1} is an eigenvalue of A^{-1} and X is a corresponding eigenvector.

$AX = \lambda X \Rightarrow A^{-1}AX = A^{-1}\lambda X \Rightarrow X = \lambda A^{-1}X \Rightarrow \lambda^{-1}X = A^{-1}X$ (since A is nonsingular, $\lambda \neq 0$).

6. [10 pts.] Find the volume of the parallelepiped with vertex at the origin and edges $U = (2, -1, 0)$, $V = (1, -2, -2)$ and $W = (3, -1, 1)$.

$V = \text{abs} \begin{vmatrix} 2 & 1 & 3 \\ -1 & -2 & -1 \\ 0 & -2 & 1 \end{vmatrix} = \text{1}$.

7. [5 + 5 = 10 pts.] Let $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 3 & 2 & 0 & 0 & 0 \\ 1 & 5 & 3 & 0 & 0 \\ 4 & 6 & 7 & 4 & 0 \\ 9 & 2 & 1 & 2 & 5 \end{bmatrix}$.

(a) Find the cofactor A_{11} of the element $a_{11} = 1$.

$$A_{11} = (1)(2)(3)(4)(5) = 120.$$

(b) Find the eigenvalues of A^{-1} .

The eigenvalues of A^{-1} are: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$.

8. [10 pts.] Find parametric equations for the line in $\mathbb{R}^3$ passing through the point $P(-2, 5, -3)$ and orthogonal to the plane $2x - 3y + 4z + 7 = 0$.

$$x = -2 + 2t, y = 5 - 3t, z = -3 + 4t, \text{ where } t \in \mathbb{R}.$$

9. [5 + 10 + 5 = 20 pts.] Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 3 & -1 \\ 3 & 0 & 0 \end{bmatrix}$.

(a) Show that the eigenvalues of A are $\lambda_1 = \lambda_2 = 3, \lambda_3 = -2$.

$$f(\lambda) = \begin{vmatrix} \lambda - 1 & 0 & -2 \\ -1 & \lambda - 3 & 1 \\ -3 & 0 & \lambda \end{vmatrix} = (\lambda - 3) \begin{vmatrix} \lambda - 1 & -2 \\ -3 & \lambda \end{vmatrix} = (\lambda - 3)(\lambda^2 - \lambda - 6) = (\lambda - 3)^2(\lambda + 2).$$

(b) For each eigenvalue λ , find the **rank** of the matrix $(\lambda I_3 - A)$.

For $\lambda_1 = \lambda_2 = 3$: $(3I_3 - A) = \begin{bmatrix} 2 & 0 & -2 \\ -1 & 0 & 1 \\ -3 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. The **rank** of $(3I_3 - A)$ is $\boxed{1}$.

Every solution to $(3I_3 - A)X = 0$ is of the form $X = \begin{bmatrix} t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, s, t \in \mathbb{R}$.

The eigenvectors of A associated with the eigenvalue 3 are of the form: $s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, where s and t are not both zero.

For $\lambda_3 = -2$: $(-2I_3 - A) = \begin{bmatrix} -3 & 0 & -2 \\ -1 & -5 & 1 \\ -3 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2/3 \\ 0 & 1 & -1/3 \\ 0 & 0 & 0 \end{bmatrix}$. The **rank** of $(-2I_3 - A)$ is $\boxed{2}$.

Every solution to $(-2I_3 - A)X = 0$ is of the form $X = t \begin{bmatrix} -2/3 \\ 1/3 \\ 1 \end{bmatrix}, t \in \mathbb{R}$.

The eigenvectors of A associated with the eigenvalue -2 are of the form: $t \begin{bmatrix} -2/3 \\ 1/3 \\ 1 \end{bmatrix}$, where $t \neq 0$.

(c) If possible, find a nonsingular matrix P and a diagonal matrix D such that $P^{-1}AP = D$.

$$P = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 3 \end{bmatrix} \text{ and } D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$